Informativeness and the Distinction Between 'Basic-Level' vs. Superordinate Verbs

Ruidi Huang, Isabelle Chapman, Anna Papafragou
Department of Linguistics, University of Pennsylvania

Language and Cognition Lab

The Basic-Level Bias in Object Naming

"animal"	"cat"	"Korat"
superordinate	basic-level Preferred	subordinate

Why is there a basic-level bias?

Conceptual organization (Murphy, 2002; Rosch, 1973; Rosch et al., 1976)

Pragmatic expectation (Choe & Papafragou, 2023)

Informativeness

- Speakers are expected to provide as much information as is required for the current communicative purpose, but no more than is necessary (Grice, 1975).
- Superordinates are generally underinformative in common contexts.

A Basic-Level Bias in Verbs?

Strict entailment

move **run**

Basic level is preferred (Zhuang & Lingnau, 2022; Zhuang et al., 2023).

However, verbs form looser hierarchies than nouns.

Partial entailment

work ←?→ **teach**

Speakers must still choose between verbs of varying specificity (even if specificity is not defined over logical scales). These choices can still be evaluated relative to pragmatic expectations of informativeness (Horn, 1972).

Does the basic-level bias extend to verbs for both types of relations?

Is this bias driven by pragmatic expectations of informativeness?

Exp. 1: Pragmatic Acceptability of "Basic" vs. Superordinate Verbs

Do adults penalize Mr. Lion's use of superordinate verbs?

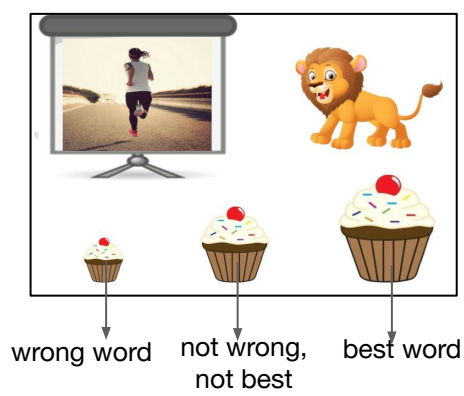
Basic-level Verb

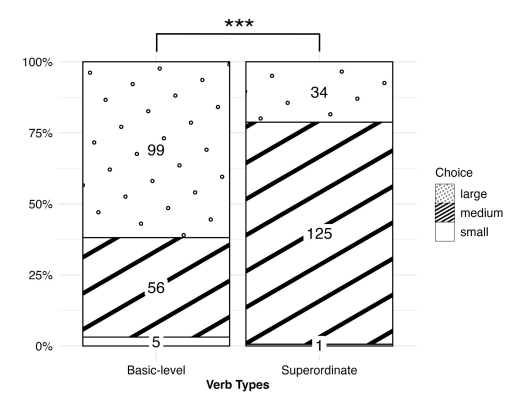
A woman is running.

Superordinate Verb

A woman is moving.

Task: Reward Mr. Lion with a cupcake





The "basic-level" bias extends to verbs.

Exp. 2: Can Context Change Acceptability of Superordinate Verbs?

Same task but different visual scene

Experimenter: "Hey, Mr. Lion, this one is different! What is it about?"

Basic-level Verb

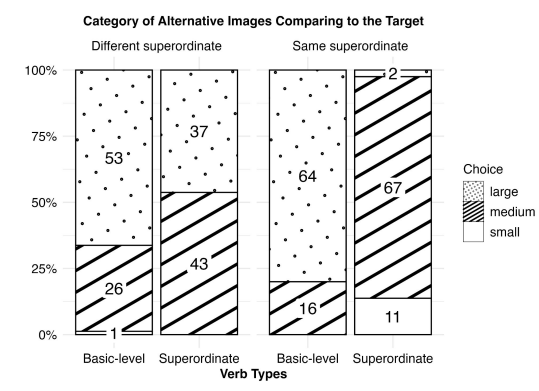
A woman is running.

Superordinate Verb

A woman is moving.

Different-superordinate alternatives

Same-superordinate alternatives



The basic-level bias in verbs is highly flexible: Superordinate verbs became significantly more acceptable when they provided enough information to pick one scene among visual alternatives.

Conclusions

- The basic-level advantage extends to verbs.
- The bias is not merely a reflection of stable conceptual hierarchies,
  - but is dynamically sensitive to the pragmatic requirement of being "informative enough" for a communicative goal.

References

Choe, J., & Papafragou, A. (2023). The acquisition of subordinate nouns as pragmatic inference. *Journal of Memory and Language*, *132*, 104432. **Grice, H. P. (1975).** Logic and conversation. In *Speech acts* (pp. 41-58). Brill. **Murphy, G. L. (2002).** *The Big Book of Concepts*. The MIT Press. **Rosch, E. H. (1973).** Natural categories. *Cognitive Psychology*, *4*(3), 328–350. **Rosch, E., Mervis, C. B., Gray, W. D., Johnson, D. M., & Boyes-Braem, P. (1976).** Basic objects in natural categories. *Cognitive Psychology*, *8*(3), 382–439. **Zhuang, T., Kabulska, Z., & Lingnau, A. (2023).** The Representation of Observed Actions at the Subordinate, Basic, and Superordinate Level. *The Journal of Neuroscience*, *43*(48), 8219–8230. **Zhuang, T., & Lingnau, A. (2022).** The characterization of actions at the superordinate, basic, and subordinate levels. *Psychological Research*, *86*(6), 1871–1891.